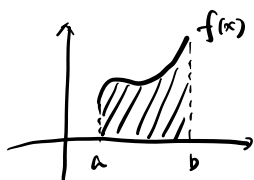


Lecture 11 Double Integrals

Recap $\int_a^b f(x) dx$



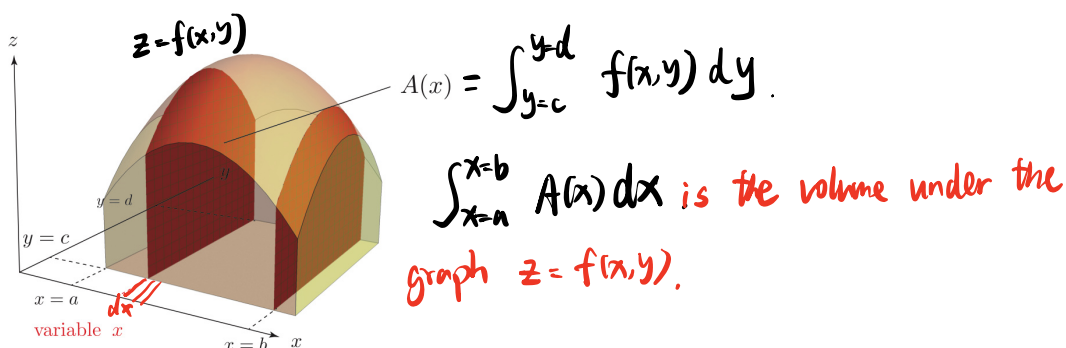
Example. Compute the following double integral:

$$\underbrace{\int_{y=1}^{y=2} \underbrace{\int_{x=0}^{x=1} (4-x-y^2x) dx}_{\text{inner}} dy}_{\text{outer}},$$

$$\begin{aligned} \text{Sol. } A(y) &= \int_{x=0}^{x=1} (4-x-y^2x) dx \\ &= \left[4x - \frac{x^2}{2} - \frac{y^2 x^2}{2} \right]_{x=0}^{x=1} \\ &= \frac{7}{2} - \frac{y^2}{2} \end{aligned}$$

$$\begin{aligned} \int_{y=1}^{y=2} A(y) dy &= \int_{y=1}^{y=2} \left(\frac{7}{2} - \frac{y^2}{2} \right) dy = \left[\frac{7}{2}y - \frac{y^3}{6} \right]_{y=1}^{y=2} \\ &= \left(7 - \frac{4}{3} \right) - \left(\frac{7}{2} - \frac{1}{6} \right) = \frac{7}{3} \end{aligned}$$

$$\begin{aligned} \int_{x=0}^{x=1} \int_{y=1}^{y=2} (4-x-y^2x) dy dx &= \int_{x=0}^{x=1} \left[4y - xy - \frac{y^3x}{3} \right]_{y=1}^{y=2} dx \\ &= \int_{x=0}^{x=1} \left[\left(8 - 2x - \frac{8x}{3} \right) - \left(4 - x - \frac{x}{3} \right) \right] dx \\ &= \int_{x=0}^{x=1} \left(4 - \frac{10x}{3} \right) dx \\ &= \left[4x - \frac{5x^2}{3} \right]_{x=0}^{x=1} \\ &= \frac{7}{3} \end{aligned}$$



Theorem (Fubini's Theorem for Rectangular Regions) Let $f(x, y)$ be a continuous function over a rectangular region $a \leq x \leq b$ and $c \leq y \leq d$, then

$$\int_{y=c}^{y=d} \int_{x=a}^{x=b} f(x, y) dx dy = \int_{x=a}^{x=b} \int_{y=c}^{y=d} f(x, y) dy dx.$$

Notation: $\int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx = \iint_R f(x, y) dA$

Example. Evaluate $\iint_R y e^{xy} dA$, where $R = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq \ln 2\}$

Sol. $\int_0^1 \int_0^{\ln 2} y e^{xy} dy dx$ integration by parts.

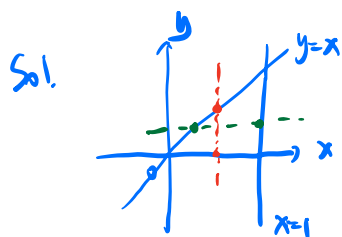
$$\begin{aligned} \int_0^{\ln 2} \int_0^1 y e^{xy} dx dy &= \int_0^{\ln 2} \int_0^1 e^{xy} d(xy) dy \\ &= \int_0^{\ln 2} e^{xy} \Big|_{x=0}^{x=1} dy \end{aligned}$$

$$= \int_0^{\ln 2} (e^y - 1) dy$$

$$= [e^y - y] \Big|_0^{\ln 2}$$

$$= 1 - \ln 2.$$

Example. Find the volume of the solid under the plane $z = 3 - x - y$ over the triangle region R bounded by the x -axis, $x=1$, $y=x$.



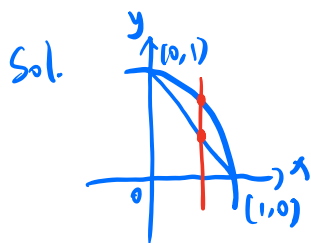
$$\begin{aligned}
 & \iint_R (3-x-y) dA \\
 &= \int_{x=0}^{x=1} \int_{y=0}^{y=x} (3-x-y) dy dx \\
 &= \int_{x=0}^{x=1} \left[3y - xy - \frac{y^2}{2} \right]_{y=0}^{y=x} dx \\
 &= \int_{x=0}^{x=1} \left(3x - \frac{3x^2}{2} \right) dx \\
 &= \left[\frac{3x^2}{2} - \frac{x^3}{2} \right]_0^1 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \iint_R (3-x-y) dA &= \int_0^1 \int_y^1 (3-x-y) dx dy \\
 &= \int_0^1 \left[3x - \frac{1}{2}x^2 - xy \right]_{x=y}^{x=1} dy \\
 &= \int_0^1 \left(\frac{3}{2} - y - 3y + \frac{3}{2}y^2 \right) dy \\
 &= \left[\frac{3}{2}y - 2y^2 + \frac{1}{2}y^3 \right]_0^1 \\
 &= 1
 \end{aligned}$$

Theorem (Fubini's Theorem for General Regions) Let R be a region on the xy -plane and $f(x,y)$ is a continuous function on R , then

$$\iint_R f(x,y) dx dy = \iint_R f(x,y) dy dx = \iint_R f(x,y) dA.$$

Example. Let R be the region in the first quadrant of the xy -plane bounded by the unit circle $x^2 + y^2 = 1$ and the straight-line $x+y=1$. Evaluate the integral $\iint_R \sqrt{1-x^2} dA$.



$$\begin{aligned}
 & \int_0^1 \int_{1-x}^{\sqrt{1-x^2}} \sqrt{1-x^2} \, dy \, dx \\
 &= \int_0^1 \left[y \sqrt{1-x^2} \right]_{1-x}^{\sqrt{1-x^2}} dx \\
 &= \int_0^1 (1-x^2) - (1-x)\sqrt{1-x^2} \, dx \\
 &= \int_0^1 (1-x^2) dx - \int_0^1 \sqrt{1-x^2} \, dx + \int_0^1 x\sqrt{1-x^2} \, dx
 \end{aligned}$$

$$\int_0^1 (1-x^2) dx = \left[x - \frac{1}{3} x^3 \right]_0^1 = \frac{2}{3}$$

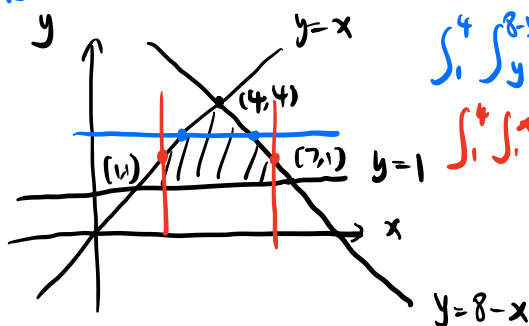
$$x = \sin \theta, \quad dx = \cos \theta \, d\theta$$

$$\int_0^1 \sqrt{1-x^2} \, dx = \int_0^{\pi/2} \cos^2 \theta \, d\theta = \int_0^{\pi/2} \frac{\cos 2\theta + 1}{2} \, d\theta = \frac{\pi}{4} + \left[\frac{1}{4} \sin 2\theta \right]_0^{\pi/2} = \frac{\pi}{4}$$

$$u = 1-x^2, \quad du = -2x \, dx$$

$$\int_0^1 x\sqrt{1-x^2} \, dx = \int_1^0 -\frac{1}{2} \sqrt{u} \, du = \int_0^1 \frac{1}{2} \sqrt{u} \, du = \left[\frac{1}{3} u^{\frac{3}{2}} \right]_0^1 = \frac{1}{3}$$

$$\Rightarrow \iint_R \sqrt{1-x^2} \, dA = 1 - \frac{\pi}{4}$$



$$\int_1^4 \int_y^{8-y} f(x,y) \, dx \, dy$$

$$\int_1^4 \int_1^x f(x,y) \, dy \, dx + \int_4^7 \int_1^{8-x} f(x,y) \, dy \, dx.$$